

ROADMAP PLANNING

Tech Club Roadmap



Options

1. Everyone works on the Q&A website
2. Everyone works on different pages for a common tech club website
3. Students work on different projects



1. Daraja Q&A

1. Everyone works on the QnA website

- Idea: All students work on different parts of the QnA website as envisioned by Vivian, who will lead the project.
- Example: Some students style the page, some work on the database of questions and answers, some work on functionality of buttons, etc
- Pros:
 - Everyone contributes → faster progress
 - Reasonable to buy one domain
 - Experience with collaborative software development
- Cons:
 - Unclear how to contribute → admin work for Vivian
 - No room for other ideas
 - Possible chaos in Github



Plan: example

Divide into teams:

- Front-end team working in HTML / CSS (most)
 - Decide layout of website (info / FAQ / chatbot)
 - Work on different parts of layout with HTML
 - Style parts with CSS
- Database team working in TXT / *Vector Database*
 - Gather facts about Daraja and add these to a text file
 - Turn the text file into a vector database using an online tool like Chroma / Pinecone
- Back-end team working in HTML / JavaScript
 - Make scripts for the front-end team (chatbot)



Plan: example

Front-end team:

1. Simplify the current code (using AI) to only the essential parts
2. Decide the website layout on paper or in powerpoint
3. Add / delete / move HTML elements to get to the desired layout
4. Style everything
 - a. Change size of elements
 - b. Change color of text and backgrounds
 - c. Add images



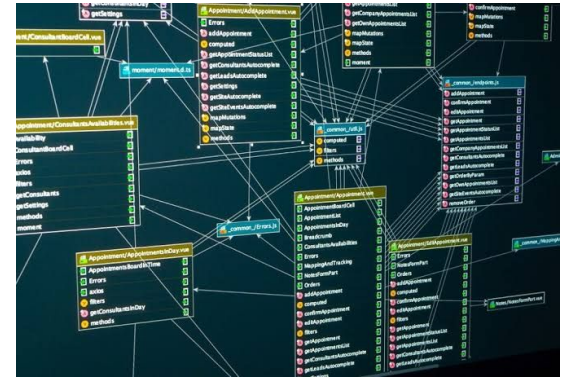
InterviewBit



Plan: example

Database team:

1. Gather facts about Daraja in a text document
2. Sort some of the data into an FAQ (in a separate document)
 - a. Q: Where is Daraja?
 - b. A: Outside Nanyuki
3. Convert the first document into a vector database (for the chatbot)
 - a. E.g. using Chroma



Plan: example

Back-end team:

1. Fix the broken google maps pane
2. Learn JavaScript through easier projects
 - a. Next two weeks
 - b. End-goal: add functionality to the message box
3. Write code for the chatbot



2. Tech Club Website

2. New tech club website

- Idea: Common tech club website that links to other pages (e.g. DarajaQnA and other personal/collaborative projects)
- Example: Some students collaborate on the QnA page, others collaborate on a Daraja News page. Some work on their own pages
- Pros:
 - One domain
 - Easy to do in github
 - New members can learn html on their own page
- Cons:
 - Less clear end goal
 - Less students per project
 - Harder to manage



Welcome to the Daraja tech club website

Todo:

- Add functionality
- Establish graphic profile
- Move domain

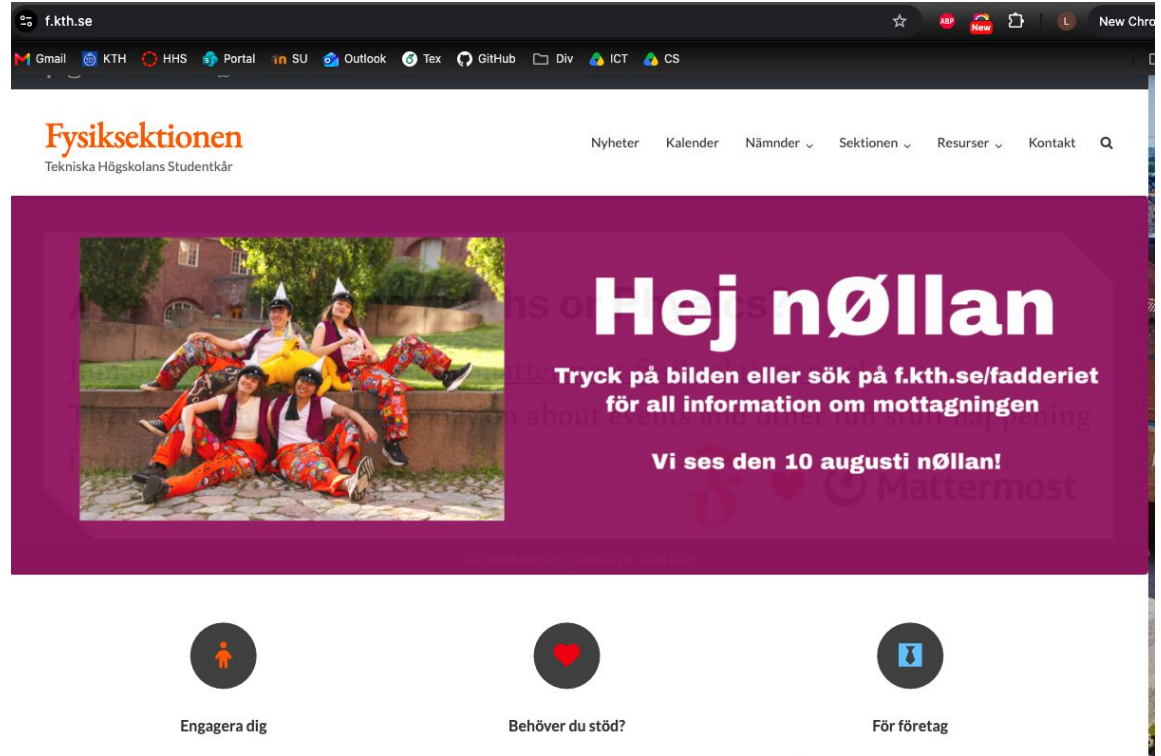
Plan: example

Divide into webpages:

- Daraja Q&A
- News website
- Project showcase
- Event calendar
- Student resource hub

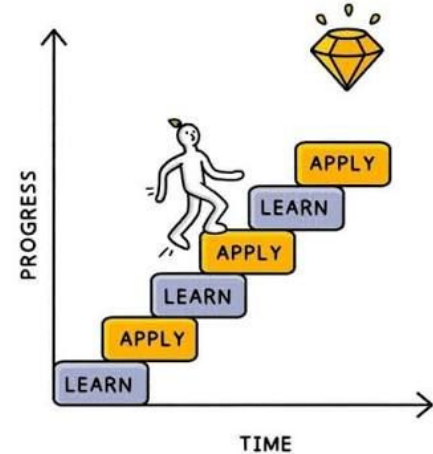
Also add personal page

- Info (bio, link to github)
- Blog
- Resume



Plan: example

1. Work on personal website while learning HTML
2. Elect a student/group responsible for main page
 - a. Link to all member pages and projects
3. Find other students interested in the same project
 - a. Formalize the idea
 - b. Make a step by step plan
 - c. Create a new subdirectory and start working
 - d. Learn with intention



3. Different projects

3. Different projects

- Idea: QnA website remains the main project. Optional to contribute. Students that prefer other projects than websites pursue those (e.g. Scratch).
- Example: Some students work on the QnA website, others work on a Daraja News website (separate github repo). Some work on a game in Scratch
- Pros:
 - More member freedom
 - No github chaos
 - Not a requirement to learn HTML
- Cons:
 - Multiple domains
 - No clear end goal
 - Much harder to manage



Plan: example

- Mix of the other two plans
- Students who do not want to do HTML can learn another tech skill
 - Programming
 - Game making in Scratch
 - Programming in Python
 - Creative
 - Digital art in Pixlr
 - Music production in Audacity
 - Video editing in CapCut
- Self learning through video tutorials
- Schedule a gallery where students can show their projects

